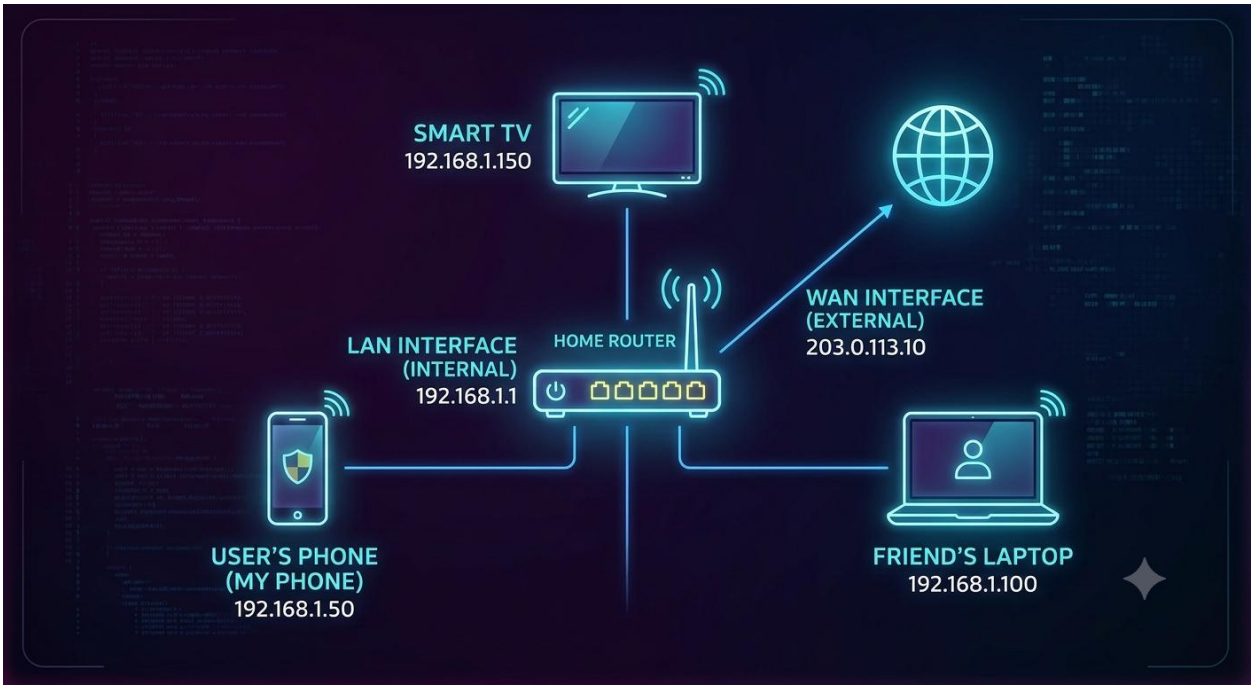
1 Draw a simple network diagram showing three devices (like your phone, a friend's laptop, and a smart TV) connected through a router, and label the IP addresses for each device and the router's interfaces.



- 2 Simulate manual IP routing by writing a step-by-step explanation of how a packet would travel from your laptop to a friend's computer in a different subnet, including

which router(s) it passes through and how the routing decision is made.

Hint: Use made-up IP addresses and show the routing table entries that would be checked at each hop.

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- **Manual IP Routing Simulation**

- **Scenario**

- My laptop sends a packet to my friend's computer, which is on a different subnet. The packet travels through a router to reach the destination.
- Network diagram
    - My Laptop
    - IP: 192.168.1.10/24
    - Router R1
    - LAN: 192.168.1.1
    - Next network: 10.0.0.1
    - Router R2
    - Next network: 192.168.2.1
    - Friend's Computer
    - IP: 192.168.2.20/24

- **Step 1: Laptop checks the destination IP**

- Source IP: `192.168.1.10`
- Destination IP: `192.168.2.20`
- My laptop checks its subnet mask (`255.255.255.0`).
- Because the destination is on a different subnet, the laptop sends the packet to its default gateway, `192.168.1.1`.

Step 2: Router R1 checks its routing table

Destination Network	Next Hop	Action
`192.168.1.0/24`	Directly connected	Local network

10.0.0.0/30	Directly connected	Transit network
192.168.2.0/24	10.0.0.2	Forward to R2
0.0.0.0/0	Other gateway	Default route

R1 finds the matching route for 192.168.2.0/24 and forwards the packet to R2 at 10.0.0.2.

Step 3: Router R2 checks its routing table

Destination Network	Next Hop	Action
10.0.0.0/30	Directly connected	Transit network
192.168.2.0/24	Directly connected	Deliver to friend's LAN
0.0.0.0/0	Other gateway	Default route

R2 finds that 192.168.2.0/24 is directly connected, so it forwards the packet to the friend's computer.

Step 4: Friend's computer receives the packet

The friend's computer receives the packet addressed to 192.168.2.20 and processes it.

Step 5: How routing decisions are made

Routers check their routing tables and select the most specific matching route (longest-prefix match). If no matching route exists, they may use a default route; otherwise, the packet is dropped.

Conclusion: The packet travels from my laptop through Router R1 and Router R2 to my friend's computer. Each router checks its routing table to decide the next ho

3 Create a routing table for a fictional Zomato-style delivery network with three branches in different cities, each on a different subnet. Specify the network address, subnet mask, next hop, and interface for each route.

Routing Table for a Zomato-Style Delivery Network

Suppose a food delivery company has three branches in different cities: Ahmedabad, Mumbai, and Delhi. Each branch uses a different subnet and communicates through routers.

Network Details

Branch	City	Network Address	Subnet Mask
Branch 1	Ahmedabad	192.168.10.0	255.255.255.0
Branch 2	Mumbai	192.168.20.0	255.255.255.0
Branch 3	Delhi	192.168.30.0	255.255.255.0

Routing Table

The following table is for the main router at the Ahmedabad branch.

Destination Network	Subnet Mask	Next Hop	Interface
192.168.10.0	255.255.255.0	Directly Connected	GigabitEthernet0/0
192.168.20.0	255.255.255.0	10.0.0.2	GigabitEthernet0/1
192.168.30.0	255.255.255.0	10.0.0.6	GigabitEthernet0/2

Explanation

1. Ahmedabad: The network is directly connected to the main router.
2. Mumbai: Packets are forwarded to the next-hop router at `10.0.0.2`.
3. Delhi: Packets are forwarded to the next-hop router at `10.0.0.6`.
4. Interfaces: The router uses the specified interface to send packets toward each destination.

Conclusion: This routing table helps the fictional food delivery company connect its three branches and route data to the correct city network. The IP addresses and interfaces are examples for learning purposes.

4 Compare static routing and dynamic routing protocols (like RIP or OSPF) by listing 3 pros and 3 cons of each, using examples from real-world apps or services you use (e.g., how a food delivery app might benefit from dynamic routing).

Comparison: Static Routing vs Dynamic Routing

Feature	Static Routing	Dynamic Routing (RIP/OSPF)
Meaning	Routes are configured manually by an administrator.	Routers learn and update routes automatically using routing protocols.
Example	A small office router uses a fixed route to reach its server.	A food delivery service such as Zomato uses a network where routers can adapt to link failures.

1. Static Routing

Pros:

1. Simple for small networks: Easy to configure in a small office.
2. More control: The administrator decides the exact route.
3. Low overhead: It does not exchange routing updates, saving bandwidth and router resources.

Cons:

1. Manual updates: Routes must be changed by an administrator when the network changes.
2. Poor scalability: Managing many routes across multiple branches becomes difficult.
3. No automatic recovery: If a route fails, an alternative route must be configured or already exist.

Real-world example: A small restaurant uses a fixed route to connect its billing system to a local server.

2. Dynamic Routing (RIP/OSPF)

Pros:

1. Automatic updates: Routers learn routes and update them when network conditions change.
2. Better scalability: Suitable for networks connecting multiple branches in different cities.
3. Adaptability: Can select another available route when a network link fails.

Cons:

1. More complexity: Requires protocol configuration and troubleshooting.
2. Resource usage: Routing updates consume some bandwidth, CPU, and memory.
3. Convergence delay: It may take time for all routers to learn about a network change.

Real-world example: A food delivery company with branches in Ahmedabad, Mumbai, and Delhi could use OSPF so routers can find alternate paths if a link between branches fails. This supports the network infrastructure behind services such as order processing and restaurant communication.

Conclusion :

Static routing is best for small, stable networks where routes rarely change. Dynamic routing, such as RIP or OSPF, is more suitable for larger networks that need automatic route updates and recovery from link failures.

5 Use ChatGPT or another AI tool to generate a sample configuration for setting up OSPF routing between two routers, then explain in your own words what each configuration line does.

Hint: Paste the AI-generated config and annotate each part with a short description.

Practical: OSPF Configuration Between Two Routers

1. Network Topology

Router R1

LAN: 192.168.1.0/24

Interface IP: 10.0.0.1

Network link: 10.0.0.0/30

Router R2

Interface IP: 10.0.0.2

LAN: 192.168.2.0/24

2. AI-Generated Configuration for Router R1

```
enable
configure terminal
router ospf 1
router-id 1.1.1.1
network 10.0.0.0 0.0.0.3 area 0
network 192.168.1.0 0.0.0.255 area 0
end
write memory
```

Explanation of each command

Command	Explanation

`enable`	Enters privileged EXEC mode.
`configure terminal`	Opens global configuration mode.
`router ospf 1`	Enables OSPF process 1.
`router-id 1.1.1.1`	Sets the unique OSPF router ID for R1.
`network 10.0.0.0 0.0.0.3 area 0`	Enables OSPF on the link between R1 and R2 in Area 0.
`network 192.168.1.0 0.0.0.255 area 0`	Advertises R1's LAN network through OSPF.
`end`	Exits configuration mode.
`write memory`	Saves the configuration.

3. AI-Generated Configuration for Router R2

```
enable
configure terminal
router ospf 1
router-id 2.2.2.2
network 10.0.0.0 0.0.0.3 area 0
network 192.168.2.0 0.0.0.255 area 0
end
write memory
```

Explanation

Command	Explanation

`enable`	Enters privileged EXEC mode.
`configure terminal`	Opens global configuration mode.
`router ospf 1`	Enables OSPF process 1.
`router-id 2.2.2.2`	Sets the unique OSPF router ID for R2.
`network 10.0.0.0 0.0.0.3 area 0`	Enables OSPF on the link between the two routers.
`network 192.168.2.0 0.0.0.255 area 0`	Advertises R2's LAN network through OSPF.
`end`	Exits configuration mode.
`write memory`	Saves the configuration.

4. Verify the OSPF Configuration

Run these commands on both routers:

```
show ip ospf neighbor
show ip route ospf
show ip protocols
```

- `show ip ospf neighbor` — checks whether the routers have formed an OSPF neighbor relationship.
- `show ip route ospf` — displays routes learned through OSPF.

- `show ip protocols` — displays routing protocol information.

5. Conclusion

OSPF is a dynamic routing protocol that allows routers to exchange network information and learn routes automatically. In this example, R1 and R2 use Area 0 to exchange routing information and communicate between their LANs.